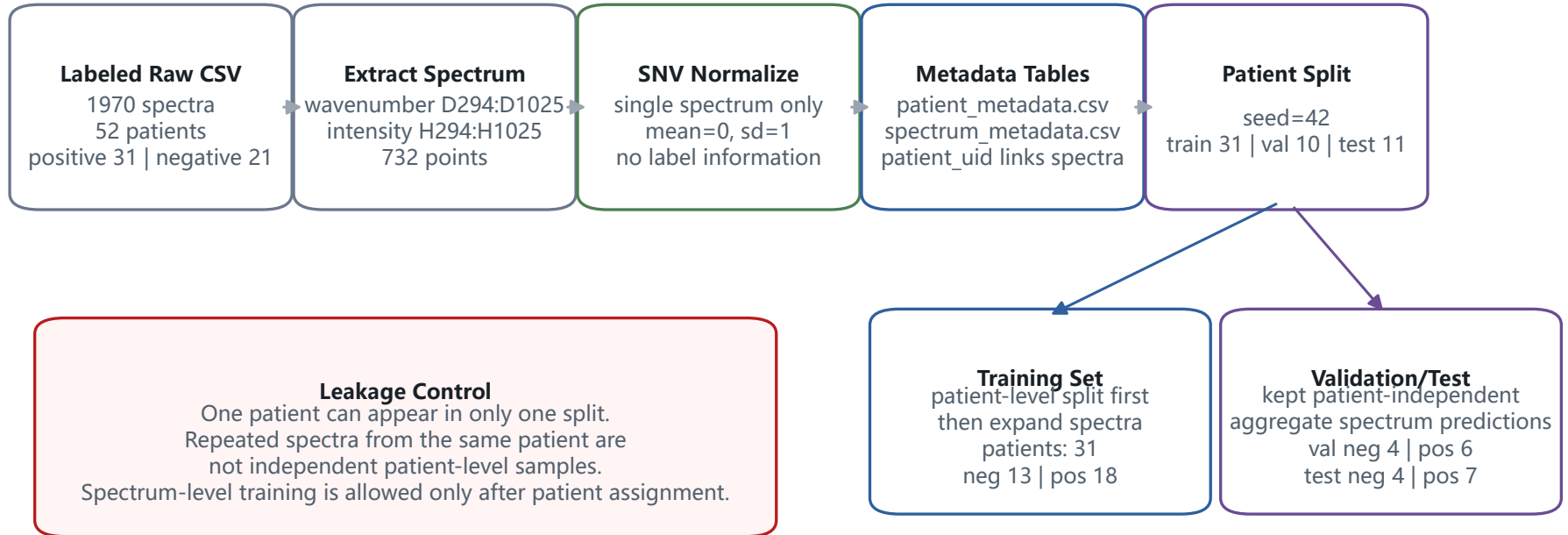


Figure 1. Phase 3A Dataset Construction Workflow

Data source: raw CSV files under DentalPlaque_SERS_Toolbox/牙菌斑SERS光谱/阳性+ and 牙菌斑SERS光谱/阴性-

Excluded from supervised Phase 3A: 未知 and 其它数据 folders. Labels are patient labels: positive=1, negative=0.



Outputs: Results/Phase3/dataset/spectra.npz, wavenumber.npy, patient_metadata.csv, spectrum_metadata.csv; Results/Phase3/splits/split_seed42.json

Verification: train/val/test patient overlap = 0; total unique split patients = 52.